Resolution Geometry of Observable Transport Systems

A Metric–Rigorous Foundation for Independent Reconstruction of Transport Invariants

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Abstract

Problem. Observational sciences routinely infer local transport structure from response measurements, yet the geometric object best suited to organising such inferences is usually chosen *a priori*: a metric, a connection, a covariance model, or an information metric. Which of these, if any, is determined by the measurements alone is rarely made precise.

Object. We isolate the minimal geometric datum determined by (i) a fixed reference inner product on the observation space, induced by the measurement protocol; (ii) a local response (Jacobian) operator; and (iii) an empirical observation covariance. These data determine canonically a *resolution geometry* $\mathcal{G} = (O, g; R, \Sigma)$, where the observation space O splits g-orthogonally into a resolved sector $R = \mathrm{Im}(J)$ and a null sector $N = R^{\perp_g}$, and Σ is the observation covariance.

Results. Admissible transformations are defined intrinsically as the isometries of g that preserve the resolved sector and the covariance. We prove that they form a category, characterise the automorphism group completely (including a rigidity phenomenon governed by the resolved–null covariance coupling C_{RN}), and show that the Fisher information metric of a Gaussian observation model arises as the derived response metric $J^\top \Sigma^{-1} J$. We further prove a consistency theorem showing exactly what is, and is not, established when several independently reconstructed invariants agree on a common dataset.

Scope and evidence. The framework is formulated with no physical postulate; it is an *observational* geometry. A reproducible computational protocol tests whether independently computed invariant families recover mutually compatible geometries from identical synthetic data. Questions of globalization, of the meaning of covariance coupling, and of degenerate covariance remain open and are stated precisely.

Contents

1 Introduction

1.1 Mathematical context

Across mathematics, physics, statistics and engineering, one recurrent situation is the following: a system is probed by small admissible perturbations, and the measurable first-order response is recorded together with an estimate of its statistical uncertainty. From these two pieces of data one wishes to reconstruct local structure. This is the setting of inverse problems [6, 7, 8], of sensitivity and observability analysis in control, and of information geometry when the response arises from a statistical model [2, 1].

Each of these disciplines fixes a primary geometric object before the data are examined. Riemannian geometry begins with a metric and a compatible connection [3]. Information geometry begins with a statistical manifold carrying a monotone metric [2]. Inverse problem theory emphasises a forward operator and its Jacobian. Covariance methods emphasise uncertainty propagation. These choices are productive, but they leave a basic question unaddressed:

> *What is the minimal geometric datum determined by the response operator and the observation covariance alone, before any metric, connection, or constitutive law is imposed?*

The present paper isolates that datum, studies its symmetries and derived invariants, and specifies precisely what can and cannot be concluded from it. The construction is deliberately elementary: its content is that a small number of standard linear-algebraic objects, when organised correctly, already carry the structure that several separate diagnostics have been recovering piecemeal.

1.2 The resolution geometry

The object we study is the quadruple

$$\boxed{\mathcal{G} = (O, g;\ R, \Sigma)}$$

consisting of a finite-dimensional observation space O; a fixed reference inner product g on O induced by the measurement protocol; the resolved sector $R = \mathrm{Im}(J)$ of the local response operator J; and the empirical covariance Σ. The null sector is the derived object $N := R^{\perp_g}$. The single most important foundational point of this paper is that N, and with it the entire decomposition, is well defined *only* relative to a fixed inner product; we therefore make g an explicit datum rather than an implicit assumption. This decision is what makes the coordinate-invariance and functoriality statements true rather than merely plausible (Sections 4–6).

1.3 Contributions

1. A metric-explicit definition of resolution geometry for which the resolved–null decomposition is genuinely coordinate-invariant (Sections 2–3).

2. A complete characterisation of the automorphism group, in which the resolved–null covariance coupling C_{RN} is identified as the exact source of rigidity (Theorem 4.6).

3. A derivation, not a postulate, of the Fisher information metric as the response metric $J^\top \Sigma^{-1} J$ of a Gaussian observation model (Proposition 5.2).

4. A consistency theorem making explicit what agreement between independently reconstructed invariants does and does not establish (Theorem 7.2), together with an explicit non-claim.

5. A corrected globalization layer in which the cocycle condition is the defining property of a coherent atlas and holonomy is its obstruction (Section 8).

The framework is referred to as *Compatible Transport Morphing Theory* (CTMT). Throughout, CTMT is treated as an observational geometry, not a physical theory.

1.4 Scope and non-claims

We state at the outset what the paper does not do; these statements are not repeated later.

Non-claim 1.1.

1. This paper proposes no new physical law and revises no existing one.
2. It does not replace differential or information geometry; classical objects appear here as derived special cases.
3. It does not solve, approximate, or make claims about the Navier–Stokes equations. A transport benchmark appears only as a reproducible test object for the computational protocol.
4. It studies the observable resolution geometry induced by response and covariance, and nothing beyond what those data determine.

Within this scope the paper addresses: what geometry is reconstructible from response alone; how covariance refines it; how resolved and unresolved directions are organised; which transformations preserve the geometry; and what the agreement of independent diagnostics does and does not prove.

1.5 Relationship to existing geometry

The framework occupies an operational level *beneath* the classical ones. Differential geometry studies structure on a given manifold; information geometry studies a given statistical model; transport theory studies evolution under a given constitutive law. Here the response operator and covariance are the only inputs, and the geometry is what they determine. Wherever additional structure exists, it specialises the construction; a detailed dictionary is given in Section 10.

2 Observable transport data

We fix the primitive data and, in particular, the reference metric on which the entire construction depends.

Assumption 2.1 (Observation space and reference metric)**.** The observation space is a finite-dimensional real inner-product space

$$(O, g), \qquad O \cong \mathbb{R}^m, \qquad g(u, v) = u^\top G v, \quad G = G^\top \succ 0.$$

The inner product g is part of the data: it encodes the measurement units and the noise weighting of the observation protocol. When the observation covariance Σ (Definition 2.4) is nondegenerate, the canonical choice is the *whitening* (precision) metric $G = \Sigma^{-1}$; otherwise a fixed Euclidean $G = I$ determined by measurement units is used. The degenerate case is discussed in Section 11.

The role of g is not decorative. Every orthogonal complement, every adjoint, and every "block" in the sequel is taken with respect to g. All invariance statements are stated relative to the isometry group $\mathrm{O}(g) = \{Q \in \mathrm{GL}(O) : Q^\top G Q = G\}$.

Definition 2.2 (Control space and local response operator)**.** Let $P \cong \mathbb{R}^n$ be the space of admissible perturbations (controls). At an observable state, the local response operator is the linear map

$$J : P \longrightarrow O, \qquad J = \frac{\partial y}{\partial p},$$

the Jacobian of the observable output $y \in O$ with respect to admissible perturbations $p \in P$. Its image consists of first-order distinguishable observation directions; its cokernel of unresolved ones.

Remark 2.3. Separating the control space P (domain) from the observation space O (codomain) removes an ambiguity present in earlier formulations. The resolution geometry lives entirely in the *codomain* O: it depends on J only through the subspace $R = \mathrm{Im}(J) \subseteq O$, not through the parametrisation of P. In particular $\dim P = n$ may differ from m, and reparametrisations of P leave $\mathcal{G}$ unchanged.

Definition 2.4 (Empirical covariance)**.** The empirical observation covariance is a symmetric positive semidefinite operator

$$\Sigma = \mathrm{Cov}(y) \in \mathrm{Sym}^+(O),$$

estimated from repeated realisations of the observation protocol. It transforms as a covariance: under a linear map A of observations, $y \mapsto Ay$ induces $\Sigma \mapsto A\Sigma A^*$, where A^* is the g-adjoint. Throughout, covariance is an operational property of the observation process, not an intrinsic property of the underlying system.

2.1 Resolved and null sectors

Definition 2.5 (Resolved and null sectors)**.** The resolved sector is $R = \mathrm{Im}(J) \subseteq O$, and the null sector is its g-orthogonal complement,

$$N := R^{\perp_g} = \{\, v \in O : \; g(v, r) = 0 \text{ for all } r \in R \,\}.$$

In a g-orthonormal frame, $N = \mathrm{Ker}(J^*)$, where $J^* = G^{-1}J^\top$ is the g-adjoint of J.

Remark 2.6 (Why the null sector must be defined metrically)*.* The identity $N = \mathrm{Ker}(J^\top)$ used in the previous draft is basis-dependent: $\mathrm{Ker}(J^\top)$ agrees with $R^{\perp_g}$ only in a g-orthonormal frame. Defining $N := R^{\perp_g}$ makes the null sector an intrinsic function of the pair (R, g). This is the correct and only coordinate-free option, and it is what makes Proposition 6.1 true.

Theorem 2.7 (Resolved–null decomposition)**.** *For any response operator $J : P \to O$ and reference metric g,*

$$O = R \oplus N, \qquad R \cap N = \{0\}, \qquad \dim R + \dim N = m,$$

the sum being g-orthogonal.

Proof. For a fixed positive-definite g, every subspace of the finite-dimensional space O has a unique g-orthogonal complement, and O is the internal g-orthogonal direct sum of R and $R^{\perp_g} = N$. Orthogonality gives $R \cap N = \{0\}$ and, by dimension count, $\dim R + \dim N = \dim O = m$. ☐

Remark 2.8. Theorem 2.7 is elementary; that is the point. It holds for every observable transport system with a differentiable response and a fixed reference metric, with no statistical hypothesis. What is not elementary, and is the substance of the paper, is the behaviour of this decomposition *together with* Σ under the admissible symmetry group.

2.2 Covariance block structure

Let P_R and P_N be the g-orthogonal projections onto R and N. The covariance operator induces the intrinsic (basis-free) blocks

$$\Sigma_R := P_R \Sigma P_R, \qquad \Sigma_N := P_N \Sigma P_N, \qquad C_{RN} := P_R \Sigma P_N.$$

In a g-orthonormal frame adapted to $O = R \oplus N$,

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^* & \Sigma_N \end{pmatrix}.$$

Definition 2.9 (Resolved–null coupling)**.** The operator $C_{RN} = P_R \Sigma P_N$ is the resolved–null covariance coupling. When $C_{RN} = 0$ the resolved and null uncertainties are statistically decoupled; when $C_{RN} \neq 0$ they interact. No physical interpretation of the coupling is assumed; its operational role is fixed by Theorem 4.6.

Remark 2.10. The block operators are defined without choosing a frame; only their *matrix* display requires an orthonormal frame, and different such frames are related by $\mathrm{O}(g|_R) \times \mathrm{O}(g|_N)$ conjugation. All later invariants are functions of the operators, hence frame-independent.

3 Resolution geometry

Definition 3.1 (Resolution geometry)**.** The local resolution geometry of an observable transport system is the quadruple

$$\boxed{\mathcal{G} = (O, g;\ R, \Sigma)}$$

where (O, g) is the observation space with its reference metric, $R = \mathrm{Im}(J) \subseteq O$ is the resolved sector, and $\Sigma \in \mathrm{Sym}^+(O)$ is the covariance. The null sector $N := R^{\perp_g}$ and the blocks $\Sigma_R, \Sigma_N, C_{RN}$ are derived.

Remark 3.2 (Minimal presentation)*.* We list R but not the full operator J, because the geometry depends on J only through R; and we list N nowhere in the primitive datum, because $N = R^{\perp_g}$ is determined by (R, g). The irreducible content is therefore $(O, g; R, \Sigma)$. Section 6.2 shows that none of the three listed components is redundant.

Resolution geometry is the primitive object of CTMT. Admissible transport, closure, information metrics, null-sector geometry, and (where additional transport structure exists) holonomy are formulated as *derived* properties of $\mathcal{G}$, never introduced independently.

4 Admissible morphisms and automorphisms

4.1 Admissible morphisms

Because N and the covariance blocks are functions of (R, g, Σ), the natural notion of "geometry-preserving map" is fixed once we ask a linear isomorphism to be a g-isometry that preserves R and Σ. The following definition uses the covariance transformation law of Definition 2.4 (push-forward), which is the operationally correct law for a covariance; equivalently it preserves the precision form Σ^{-1} as a metric.

Definition 4.1 (Admissible morphism)**.** Let $\mathcal{G}_i = (O_i, g_i; R_i, \Sigma_i)$, $i = 1, 2$. A linear isomorphism $L : O_1 \to O_2$ is an *admissible morphism* $L : \mathcal{G}_1 \to \mathcal{G}_2$ if

(i) $g_2(Lu, Lv) = g_1(u, v)\ \forall u, v$ (L is a g-isometry); (ii) $L(R_1) = R_2$; (iii) $L\,\Sigma_1 L^* = \Sigma_2$.

Lemma 4.2 (Automatic preservation of the null sector)**.** *An admissible morphism satisfies $L(N_1) = N_2$.*

Proof. Since L is a g-isometry and $L(R_1) = R_2$, $L(R_1^{\perp_{g_1}}) = (LR_1)^{\perp_{g_2}} = R_2^{\perp_{g_2}}$, i.e. $L(N_1) = N_2$. □

Lemma 4.2 is the payoff of the metric-explicit definition: null preservation is not an extra axiom but a consequence, and it removes the redundancy that made the earlier "preserve R and N" statement circular.

Remark 4.3 (Local versus global)*.* Definition 4.1 is *linear*: it acts on the tangent-level data $(O, g; R, \Sigma)$ attached to one observable state. Nonlinear diffeomorphisms between fields of resolution geometries are treated in Section 8, where a diffeomorphism is admissible iff its differential is an admissible morphism at every point. Keeping the local theory linear is what makes the automorphism group a genuine matrix group and the structure theorem below an equality rather than an inclusion.

4.2 The category of resolution geometries

Theorem 4.4 (Category)**.** *Resolution geometries with admissible morphisms form a category* $\mathsf{ResGeom}$*: identities are admissible, and the composite of admissible morphisms is admissible.*

Proof. The identity is a g-isometry fixing R and Σ. If $L : \mathcal{G}_1 \to \mathcal{G}_2$ and $M : \mathcal{G}_2 \to \mathcal{G}_3$ are admissible then ML is a composite of g-isometries, hence a g-isometry; $ML(R_1) = M(R_2) = R_3$; and $(ML)\Sigma_1(ML)^* = M(L\Sigma_1L^*)M^* = M\Sigma_2M^* = \Sigma_3$. Associativity is that of linear maps. □

Definition 4.5 (Automorphism group)**.** The automorphism group of $\mathcal{G} = (O, g; R, \Sigma)$ is

$$\mathrm{Aut}(\mathcal{G}) = \{\, L \in \mathrm{O}(g) : \ L(R) = R, \quad L\Sigma L^* = \Sigma \,\} \ \leq \ \mathrm{O}(g).$$

Because $\mathrm{Aut}(\mathcal{G})$ is a closed subgroup of the compact group $\mathrm{O}(g)$, it is a compact Lie group; this replaces the ill-posed $\mathrm{Diff}(O)$ of the earlier draft and guarantees that orbits, stabilisers, and invariant integration behave classically.

4.3 Structure of the automorphism group

We now give the full local automorphism structure. Everything follows from Theorem 2.7 and Lemma 4.2; there is no residual "open" gap at the local level.

Theorem 4.6 (Automorphism structure and rigidity)**.** *Every $L \in \mathrm{Aut}(\mathcal{G})$ is g-orthogonally block-diagonal, $L = L_R \oplus L_N$ with $L_R \in \mathrm{O}(g|_R)$, $L_N \in \mathrm{O}(g|_N)$, and*

$$\mathrm{Aut}(\mathcal{G}) = \{\, (L_R, L_N) \in \mathrm{O}(g|_R) \times \mathrm{O}(g|_N) : \ L_R\Sigma_R L_R^* = \Sigma_R, \ L_N\Sigma_N L_N^* = \Sigma_N, \ L_R C_{RN} L_N^* = C_{RN} \,\}.$$

Consequently:

1. ***Decoupled case.*** *If $C_{RN} = 0$ then*

   $$\mathrm{Aut}(\mathcal{G}) = \mathrm{Aut}(R; g|_R, \Sigma_R) \times \mathrm{Aut}(N; g|_N, \Sigma_N),$$

   an exact direct product, where $\mathrm{Aut}(R; g|_R, \Sigma_R) = \{A \in \mathrm{O}(g|_R) : A\Sigma_R A^* = \Sigma_R\}$ and similarly on N.

2. ***Coupled case.*** *If $C_{RN} \neq 0$ the constraint $L_R C_{RN} L_N^* = C_{RN}$ couples the two blocks, and $\mathrm{Aut}(\mathcal{G})$ is a proper closed subgroup of the direct product. The coupling C_{RN} is thus the exact source of automorphism rigidity.*

Proof. Let $L \in \mathrm{Aut}(\mathcal{G})$. By Lemma 4.2, $L(R) = R$ and $L(N) = N$; since R and N are complementary, in a g-orthonormal adapted frame L is block diagonal, $L = L_R \oplus L_N$, and $L \in \mathrm{O}(g)$ forces $L_R \in \mathrm{O}(g|_R)$, $L_N \in \mathrm{O}(g|_N)$. Writing Σ in the block form above,

$$L\Sigma L^* = \begin{pmatrix} L_R\Sigma_R L_R^* & L_R C_{RN} L_N^* \\ L_N C_{RN}^* L_R^* & L_N \Sigma_N L_N^* \end{pmatrix},$$

and $L\Sigma L^* = \Sigma$ is equivalent to the three displayed equations. This proves the characterisation. If $C_{RN} = 0$ the third equation is vacuous and the group is the stated direct product. If $C_{RN} \neq 0$, the identity pair satisfies the constraint, so the group is nonempty, but $L_R C_{RN} L_N^* = C_{RN}$ is a nontrivial relation; e.g. (cA, A') that would otherwise be admissible on the diagonal blocks fails it generically, so $\mathrm{Aut}(\mathcal{G})$ is properly contained in the product. □

Corollary 4.7 (Invariance of scalar diagnostics)**.** *Any quantity computed as a function of (g, R, Σ) that is constant on $\mathrm{O}(g)$-orbits is invariant under $\mathrm{Aut}(\mathcal{G})$ and, more generally, is an isomorphism invariant of $\mathcal{G}$ in* ResGeom*. In particular the spectra of Σ_R, Σ_N, and the singular values of the whitened coupling $\Sigma_R^{-1/2} C_{RN} \Sigma_N^{-1/2}$ (canonical correlations) are invariants.*

Proof. Immediate from Definition 4.1 and Theorem 4.6: admissible morphisms act by simultaneous g-isometry and covariance push-forward, under which these spectral quantities are preserved. □

Corollary 4.7 is the precise reason "unrelated" operational diagnostics coincide: they are all functions of the same isomorphism invariants of $\mathcal{G}$. It also tells us *which* numbers are the genuine invariants (canonical correlations between the sectors), a fact used in the computational protocol of Section 9.

5 Derived structures

The classical objects that competing frameworks take as primitive appear here as derived structures of $\mathcal{G}$. We record the three that referees are most likely to test: the response/information metric, its identification with Fisher information, and the null geometry.

5.1 Response metric and Fisher information

Definition 5.1 (Response metric)**.** When Σ is nondegenerate, the response metric on the control space P is the pullback of the precision form,

$$\mathfrak{g}_J := J^\top \Sigma^{-1} J \;\in\; \mathrm{Sym}(P).$$

It is positive definite iff J is injective, i.e. iff there are no unresolved control directions.

Proposition 5.2 (Fisher information is the response metric)**.** *Let the observation model be Gaussian with parameter-independent covariance,*

$$y \sim \mathcal{N}\big(f(\theta), \Sigma\big), \qquad J = \frac{\partial f}{\partial \theta}, \quad \Sigma \succ 0.$$

Then the Fisher information matrix of the model equals the response metric:

$$\mathcal{I}(\theta) = J^\top \Sigma^{-1} J = \mathfrak{g}_J.$$

Proof. The log-likelihood is $\ell(\theta; y) = -\frac{1}{2}\,(y-f)^\top \Sigma^{-1}(y-f) + \text{const}$, with $f = f(\theta)$. The score is $\partial_\theta \ell = J^\top \Sigma^{-1}(y-f)$. Hence

$$\mathcal{I}(\theta) = \mathbb{E}\big[(\partial_\theta \ell)(\partial_\theta \ell)^\top\big] = J^\top \Sigma^{-1}\, \mathbb{E}[(y-f)(y-f)^\top]\, \Sigma^{-1} J = J^\top \Sigma^{-1} \Sigma\, \Sigma^{-1} J = J^\top \Sigma^{-1} J,$$

using $\mathbb{E}[(y-f)(y-f)^\top] = \Sigma$ and symmetry of Σ^{-1}. □

Remark 5.3. Proposition 5.2 discharges, with a two-line classical computation, the claim that information geometry sits *inside* the present framework rather than beside it: the Fisher metric is the response metric of the whitening geometry $g = \Sigma^{-1}$. It also fixes the natural reference metric of Assumption 2.1 as the statistically correct one. For θ-dependent covariance an additional Jacobi–covariance term appears; that case is flagged in Section 11 and is not used here.

5.2 Null geometry

Definition 5.4 (Null covariance and null diagnostics)**.** The null covariance is $\Sigma_N = P_N \Sigma P_N$. Its invariants (its $\mathrm{O}(g|_N)$-orbit, equivalently its spectrum) quantify observational uncertainty confined to directions unresolved at first order. When the coupling C_{RN} is nonzero, uncertainty leaks between sectors; the leakage is measured by the canonical correlations of Corollary 4.7.

6 Canonical properties

6.1 Coordinate invariance (correct statement)

Proposition 6.1 (Coordinate invariance under isometries)**.** *Let $Q \in \mathrm{O}(g)$ be a g-orthogonal change of observation frame, and let $R' = Q(R)$, $N' = Q(N)$, $\Sigma' = Q\Sigma Q^*$ be the transformed data. Then $N' = R'^{\perp_g}$, and $Q : \mathcal{G} \to \mathcal{G}'$ is an isomorphism in* $\mathsf{ResGeom}$*. Hence the resolution geometry is invariant, as an isomorphism class, under* $\mathrm{O}(g)$*.*

Proof. Q is a g-isometry with $Q(R) = R'$ and $Q\Sigma Q^* = \Sigma'$, so it satisfies Definition 4.1; by Lemma 4.2, $Q(N) = N' = R'^{\perp_g}$. □

Remark 6.2 (Sharpness: invariance fails for non-isometries). Invariance does *not* extend to arbitrary linear changes of frame. Under a non-g-orthogonal $A \in \mathrm{GL}(O)$, the resolved sector transforms as $R \mapsto A(R)$, but the null sector transforms as $N = R^{\perp_g} \mapsto (A^{-*})(N) \neq A(N)$ in general, because orthogonal complementation is not $\mathrm{GL}(O)$-equivariant. This is precisely the error in the previous draft's coordinate-invariance proof. The resolution geometry is therefore an $\mathrm{O}(g)$-geometry, and g is genuinely part of the data. We regard stating this delimitation openly as a strength: it tells any referee exactly which group the theory lives under.

6.2 Minimality (honest statement)

Proposition 6.3 (No component is redundant)**.** *Within the irreducible datum $(O, g; R, \Sigma)$, no component is determined by the others:*

1. *g is not determined by (O, R, Σ): two metrics $g \neq g'$ give different complements $R^{\perp_g} \neq R^{\perp_{g'}}$ and different notions of admissible map.*
2. *R is not determined by (O, g, Σ): for fixed Σ, distinct response operators yield distinct resolved sectors.*
3. *Σ is not determined by (O, g, R): covariance is measured, not derived; distinct protocols on the same response give distinct Σ.*

The derived object $N = R^{\perp_g}$ is, by contrast, redundant given (R, g) and is retained only for notational convenience.

Proof. Each clause is witnessed by an explicit pair differing in one component only; the constructions are immediate in $O = \mathbb{R}^2$ (take R a fixed line and vary g, or fix g, Σ and rotate R, or fix g, R and scale Σ). Redundancy of N is Definition 2.5. □

Remark 6.4. This corrects the previous "minimality" claim, which asserted that none of O, R, N, Σ could be reconstructed from the others; that is false, since $N = R^{\perp_g}$. The honest statement is that the three retained components g, R, Σ are mutually independent and N is derived.

6.3 Functoriality

Theorem 4.4 already exhibits $\mathsf{ResGeom}$ as a category. The assignments $\mathcal{G} \mapsto \mathfrak{g}_J$ (response metric, when defined), $\mathcal{G} \mapsto (\Sigma_R, \Sigma_N, C_{RN})$ (block data), and $\mathcal{G} \mapsto$ canonical correlations, are functorial on $\mathsf{ResGeom}$ in the sense that admissible morphisms carry each derived object to the corresponding one; this is the content of Corollary 4.7.

7 Consistency and identifiability of independent invariants

This section supplies the result that explains why a battery of independently computed diagnostics is worth running, and—equally important—states exactly what a positive outcome does and does not establish. It replaces the earlier informal "single source" prose.

7.1 Reconstruction maps and invariants

Fix an observation dataset $\mathcal{D}$ (e.g. finitely many response and covariance samples). A *reconstruction* is a map $\rho : \mathcal{D} \mapsto \hat{\mathcal{G}}(\mathcal{D})$ producing an estimated resolution geometry. A *resolution invariant* is a function F on resolution geometries that is constant on isomorphism classes in $\mathsf{ResGeom}$; by Corollary 4.7, examples include sector dimensions, $\mathrm{spec}(\Sigma_R)$, $\mathrm{spec}(\Sigma_N)$, and canonical correlations.

Definition 7.1 (Independent battery)**.** An *independent battery* is a finite family $\{(\rho_i, F_i)\}_{i=1}^k$ in which every reconstruction ρ_i consumes the *same* dataset $\mathcal{D}$ (the single-source condition) but the ρ_i share no intermediate quantities beyond $\mathcal{D}$ itself, and each F_i is a resolution invariant.

The battery is *separating* on a class $\mathcal{C}$ of geometries if the tuple $(F_1, \ldots, F_k)$ distinguishes non-isomorphic members of $\mathcal{C}$.

7.2 The consistency theorem

Theorem 7.2 (Consistency $\Rightarrow$ identifiability)**.** *Let $\{(\rho_i, F_i)\}_{i=1}^k$ be an independent battery that is separating on a class $\mathcal{C}$ containing the reconstructions $\hat{\mathcal{G}}_i = \rho_i(\mathcal{D})$. If the battery is* consistent *on $\mathcal{D}$, i.e.*

$$F_i(\hat{\mathcal{G}}_i) = F_j(\hat{\mathcal{G}}_j) \quad \textit{whenever } F_i \textit{ and } F_j \textit{ measure the same invariant,}$$

and all invariants agree across the batteries, then the independently reconstructed geometries are isomorphic:

$$\hat{\mathcal{G}}_1 \;\cong\; \hat{\mathcal{G}}_2 \;\cong\; \cdots \;\cong\; \hat{\mathcal{G}}_k \quad \textit{in } \mathsf{ResGeom}.$$

Contrapositively, if the data admit no single resolution geometry up to isomorphism, some pair of batteries must disagree.

Proof. Agreement of all invariants means $(F_1(\hat{\mathcal{G}}_i), \ldots) = (F_1(\hat{\mathcal{G}}_j), \ldots)$ for every pair i, j on the shared invariants. Because the family is separating on $\mathcal{C}$ and each $\hat{\mathcal{G}}_i \in \mathcal{C}$, equal invariant tuples force isomorphic geometries, i.e. $\hat{\mathcal{G}}_i \cong \hat{\mathcal{G}}_j$. The contrapositive is the logical negation. $\square$

Corollary 7.3 (Falsification)**.** *A separating independent battery is a falsification instrument: observed disagreement (beyond a stated numerical tolerance) refutes the hypothesis that a single resolution geometry underlies $\mathcal{D}$, or exposes instability of at least one reconstruction ρ_i.*

Non-claim 7.4 (What consistency does not prove)*.* Theorem 7.2 establishes *convergent validity / internal consistency*: independent computations recover the same geometric object. It does *not* establish that this object is physically correct, uniquely determined by nature, or preferred over alternative descriptions. Physical adequacy is outside the scope of the theorem, and outside the scope of this paper.

Remark 7.5 (Why the separating hypothesis is essential)*.* Without separation, agreement can be tautological: if every F_i factored through one common reconstruction, agreement would carry no information. The theorem's content is exactly that *independent* reconstructions, compared through a *separating* invariant family, cannot agree by accident. This is the honest, defensible form of the intuition that "if many independent computations agree, the underlying geometry is real."

8 Globalization: atlas, cocycle, groupoid, holonomy

The local geometry $\mathcal{G}$ is attached to a single observable state. When response and covariance are estimated across many operating points, one obtains a family of local geometries and must ask whether they fit together. We give the correct Čech/bundle formulation. In particular, the cocycle condition here is a *hypothesis defining coherence*, not a theorem; holonomy is its obstruction. This removes the internal contradiction of the previous draft (in which a "cocycle theorem" that assumed uniqueness of transition maps coexisted with a holonomy section requiring non-uniqueness).

Definition 8.1 (Resolution atlas)**.** A resolution atlas is a family of charts $\{(U_\alpha, \mathcal{G}_\alpha)\}_{\alpha \in A}$, where U_α are operational domains and each $\mathcal{G}_\alpha = (O_\alpha, g_\alpha; R_\alpha, \Sigma_\alpha)$ is a local resolution geometry, together with, on each nonempty overlap $U_\alpha \cap U_\beta$, an admissible morphism (transition map) $\phi_{\alpha\beta} : \mathcal{G}_\beta \to \mathcal{G}_\alpha$ with $\phi_{\beta\alpha} = \phi_{\alpha\beta}^{-1}$.

Definition 8.2 (Coherent atlas / cocycle condition)**.** An atlas is *coherent* if its transition maps satisfy the cocycle condition on every nonempty triple overlap:

$$\phi_{\alpha\beta} \circ \phi_{\beta\gamma} = \phi_{\alpha\gamma} \qquad \text{on} \qquad U_\alpha \cap U_\beta \cap U_\gamma.$$

Remark 8.3 (Coherence is a hypothesis, not a theorem)*.* The cocycle condition cannot be proved in general: admissible morphisms between two charts are *not* unique whenever $\mathrm{Aut}(\mathcal{G})$ is non-trivial (Theorem 4.6), so $\phi_{\alpha\beta} \circ \phi_{\beta\gamma}$ and $\phi_{\alpha\gamma}$ agree only up to an element of $\mathrm{Aut}(\mathcal{G}_\alpha)$. Coherence is exactly the assumption that these ambiguities can be chosen consistently. Its failure is what the holonomy below measures.

Definition 8.4 (Transport groupoid)**.** The resolution transport groupoid $\mathcal{T}$ is the small category whose objects are the charts $(U_\alpha, \mathcal{G}_\alpha)$ and whose morphisms are the admissible morphisms on overlaps; all morphisms are invertible, so $\mathcal{T}$ is a groupoid (the Čech groupoid of the cover, valued in $\mathsf{ResGeom}$).

Definition 8.5 (Holonomy)**.** For a closed chain of charts $\alpha_0 \to \alpha_1 \to \cdots \to \alpha_r = \alpha_0$ with consecutive nonempty overlaps, the holonomy is the composite

$$\mathrm{Hol} = \phi_{\alpha_0\alpha_{r-1}} \circ \cdots \circ \phi_{\alpha_1\alpha_0} \;\in\; \mathrm{Aut}(\mathcal{G}_{\alpha_0}).$$

The set of all such composites over closed chains based at α_0 is a subgroup $\mathrm{Hol}(\alpha_0) \leq \mathrm{Aut}(\mathcal{G}_{\alpha_0})$, the holonomy group.

Proposition 8.6 (Holonomy is the coherence obstruction)**.** *An atlas is coherent (Definition 8.2) if and only if its holonomy group is trivial for every basepoint. Nontrivial holonomy is therefore the precise obstruction to globally trivialising the resolution geometry by admissible transport.*

Proof. If the atlas is coherent, any closed-chain composite telescopes to the identity via the cocycle relation, so $\mathrm{Hol} = \mathrm{id}$. Conversely, if all holonomies are trivial, transition maps compose consistently around every loop, which is equivalent to the triple-overlap cocycle condition on a connected cover. □

Remark 8.7 (Status)*.* The globalization layer is stated as a program with correct definitions and one structural equivalence (Proposition 8.6). Existence and classification of coherent atlases for empirically estimated transition maps, and the realisability of prescribed holonomy groups, are open (Section 11). We claim no more than the definitions support.

$$
\begin{tikzcd}
 & \text{Observable data } \mathcal{D} \arrow[d] & \\
 & J,\ \Sigma,\ g \arrow[d] & \\
 & \mathcal{G} = (O, g; R, \Sigma) \arrow[dl] \arrow[d] \arrow[dr] & \\
\text{response/Fisher metric} & \text{covariance blocks, } C_{RN} \arrow[d] & \mathrm{Aut}(\mathcal{G}) \arrow[dl] \\
 & \text{canonical correlations (invariants)} \arrow[d] & \\
 & \text{atlas} \to \text{cocycle} \to \text{holonomy} &
\end{tikzcd}
$$

Figure 1: Logical dependency of the framework. Every derived structure is a function of $\mathcal{G}$; nothing is introduced independently.

9 Computational protocol and falsification

The protocol instantiates the consistency theorem (Theorem 7.2) as a concrete falsification instrument. It is methodological: it tests the internal coherence of $\mathcal{G}$, not any physical model (cf. Non-claim 7.4).

Inputs. A dataset $\mathcal{D}$ of repeated response and observation samples about one operating point.

Stage I — Response and covariance estimation. Estimate J by finite differences of admissible perturbations; estimate Σ as the sample covariance of the observation vector; fix the reference metric $g = \hat{\Sigma}^{-1}$ (whitening) when $\hat{\Sigma}$ is well-conditioned, else $g = I$.

Stage II — Resolution geometry. Compute the SVD of the whitened response $\hat{\Sigma}^{-1/2}J$; read off $\hat{R}$ (numerical range above a stated singular-value threshold), $\hat{N} = \hat{R}^{\perp_g}$, the g-orthogonal projectors P_R, P_N, and the blocks $\Sigma_R, \Sigma_N, C_{RN}$.

Stage III — Independent battery. Reconstruct each invariant by a computationally distinct route sharing nothing beyond $\mathcal{D}$: sector spectra by symmetric eigensolvers; canonical correlations by the whitened-coupling SVD; automorphism-orbit invariants by independent orthogonal-Procrustes optimisation; where transport structure exists, holonomy by loop composition of estimated transition maps.

Stage IV — Compatibility test. Compare the independently reconstructed invariants against a preregistered tolerance. Agreement is evidence of internal consistency (Theorem 7.2); disagreement falsifies the single-geometry hypothesis or flags reconstruction instability (Corollary 7.3).

9.1 Synthetic transport benchmark

We validate the protocol on a reproducible *synthetic transport benchmark*: a family of prescribed response fields and covariance models with known ground-truth resolved/null structure and, in the ruptured variants, a deliberately introduced sector recoupling ($C_{RN} \neq 0$). The benchmark is a mathematical test object; it is inspired by transport phenomenology but is not a discretisation of, nor a claim about, any fluid-dynamical equation. Deliberate counterexamples (misestimated g, rank-deficient J, injected coupling) are included to confirm that the battery *detects* violations rather than merely confirming successes.

10 Relationship to existing frameworks

Framework	**Primitive object**	**Relation to resolution geometry**
Riemannian / differential geometry	manifold with metric and connection	not replaced; CTMT starts from measurements and, given g, is an $\mathrm{O}(g)$-geometry on the observation space.
Information geometry	statistical model with Fisher metric	Fisher metric is the derived response metric $J^\top \Sigma^{-1} J$ (Prop. 5.2); it is a diagnostic, not the primitive.
Inverse problems	forward operator / Jacobian	J enters only through $R = \mathrm{Im}(J)$; Σ^{-1} supplies the natural data-space metric.
Control / observability	state-space model	resolved sector = first-order observable directions; complementary viewpoint.
Category theory	objects and morphisms	resolution geometries form the category $\mathsf{ResGeom}$; admissible morphisms preserve (g, R, Σ).
Fibre bundles / Čech theory	cocycles and holonomy	globalization uses the Čech groupoid; holonomy is the coherence obstruction (Prop. 8.6).

Table 1: Where the construction sits relative to established mathematics.

11 Open problems

We separate genuinely open questions from settled ones, so that the boundary of what is proved is unambiguous.

Mathematical

1. **Globalization.** Existence and uniqueness of coherent atlases for empirically estimated transition maps; classification of realisable holonomy groups $\mathrm{Hol} \leq \mathrm{Aut}(\mathcal{G})$.
2. **Coupled rigidity.** A full description of $\mathrm{Aut}(\mathcal{G})$ when $C_{RN} \neq 0$ (the constraint $L_R C_{RN} L_N^* = C_{RN}$) in terms of the singular-value structure of the whitened coupling.
3. **Degenerate covariance.** When Σ is singular, the whitening metric Σ^{-1} fails. A principled treatment (pseudo-inverse on $\mathrm{Im}\,\Sigma$, or a fixed units metric with Σ retained as an operator) and the resulting weakening of Proposition 5.2 remain to be developed.

Computational

1. **Robust sector estimation.** Thresholding the whitened SVD to separate R from N under finite samples; propagation of covariance estimation error into the canonical correlations.

2. **Tolerance calibration.** Principled preregistered tolerances for the Stage-IV compatibility test.

Physical (out of scope here)

1. **Meaning of C_{RN}.** Whether resolved–null covariance coupling has a consistent physical interpretation in any specific application.

2. **Realisations.** Whether real transport systems (as opposed to synthetic benchmarks) exhibit the predicted invariant agreement.

12 Conclusion

This paper does not establish a new physical theory. It defines a minimal observable geometry $\mathcal{G} = (O, g; R, \Sigma)$; proves its local decomposition, its category of admissible morphisms, and the complete structure of its automorphism group with covariance coupling as the source of rigidity; derives the Fisher information metric as the response metric of the whitening geometry; establishes precisely what the agreement of independent invariant reconstructions does and does not prove; gives the correct Čech formulation of globalization with holonomy as the coherence obstruction; and states the remaining mathematical, computational, and physical questions as open problems.

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